

ASSUMPTIONS BLOCK — every other sheet references these cells

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Legend: blue = given input (from brief), yellow fill = editable scenario input, black = formula

Item	Value	Unit	Source / note
TRANSACTION FACTS			
Export order value (FOB), USD	\$600,000	USD	Brief 3.1
Production cost / packing credit base, USD	\$420,000	USD	Brief 3.1
Pre-shipment (production) period, days	60	days	Brief 3.1
Credit period offered to buyer, days after shipment	90	days	Brief 3.1
Reporting exchange rate, INR per USD	83.50	INR/USD	Brief 3.1 (fixed)
Day-count convention, days per year	360	days	Brief 3.1
Advance payment price concession	5.00%	%	Brief 3.4
BANK CHARGES			
LC advising charge, USD (flat)	\$150	USD	Brief 3.2
LC confirmation charge, % per month on LC value	0.15%	%/month	Brief 3.2
Confirmation tenor, months (LC opened at order, valid to shipment = 60 days)	2	months	Assumption: 60-day production = 2 months
Negotiation / handling charge, % of bill value	0.20%	%	Brief 3.2
SWIFT / courier, USD (flat)	\$120	USD	Brief 3.2
Documentary collection (D/A) handling, % of bill value	0.15%	%	Brief 3.2
Open account remittance handling, USD (flat)	\$150	USD	Brief 3.2
Inward remittance charge (advance), USD (flat)	\$100	USD	Brief 3.2
Discrepancy fee (LC), USD per set	\$90	USD	Brief 3.2
FINANCE RATES (per year)			
Pre-shipment packing credit rate	9.00%	%	Brief 3.3
Post-shipment credit / bill discounting (INR) rate	9.50%	%	Brief 3.3
Interest subvention benefit (reduction)	3.00%	%	Brief 3.3
Foreign currency usance bill discount rate	6.50%	%	Brief 3.3 (already net)
Exporter opportunity cost of capital (PV rate)	12.00%	%	Brief 3.3
Subvented packing credit rate	6.00%	%	Formula: base – subvention
Subvented post-shipment credit rate	6.50%	%	Formula: base – subvention
DEFAULT PROBABILITIES			
Advance payment (100%)	0.00%	%	Brief 3.4
Confirmed LC at sight	0.20%	%	Brief 3.4
Usance LC 90 days (unconfirmed)	1.00%	%	Brief 3.4
Documentary collection D/A 90 days	4.00%	%	Brief 3.4

Open account 90 days	8.00%	%	Brief 3.4
TASK E SCENARIO INPUTS			
Number of discrepant document sets presented	1	sets	Task E scenario
Default probability if LC is discrepant (bank undertaking lost → buyer credit risk, open-account lev	8.00%	%	Scenario (editable)
Extra days delay while buyer waiver is sought	10	days	Scenario (editable)

TASK A — Pre-shipment financing cost (packing credit)

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Step	Item	Value	Formula (as typed)
1	Packing credit principal, USD	\$420,000	=Assumptions!B8
2	Base packing credit rate	9.00%	=Assumptions!B25
3	Interest subvention	3.00%	=Assumptions!B27
4	Subvented rate (base – subvention)	6.00%	=C6-C7
5	Production period, days	60	=Assumptions!B9
6	Day count	360	=Assumptions!B12
7	Packing credit interest, USD = Principal × subvented rate × days/360	\$4,200.00	=C5*C8*C9/C10
8	Packing credit interest, INR (× 83.50)	₹350,700	=C11*Assumptions!B11
9	Interest without subvention (for comparison), USD	\$6,300.00	=C5*C6*C9/C10
10	Subvention saving, USD	\$2,100.00	=C13-C11

Note: this cost applies to every payment term except advance payment (cash arrives before production, so no packing credit is drawn).

TASK B & C — Cost build-up, net realisation, present value and all-in cost for the five payment terms

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Row	Advance payment (100%)	Confirmed LC at sight	Usance LC 90 days (unconfirmed)	Documentary collection D/A 90 days	Open account 90 days
Special condition	5% price concession demanded by buyer	Confirmation charge applies	Bill discounted in foreign currency @ 6.5%	No bank undertaking; post-shipment credit	No documents held; post-shipment credit
Payment timing (day)	0	60	60	60	60
Cash timing basis	Advance received before production	Sight payment at shipment (day 60)	Discounted proceeds at shipment (day 60)	Credit-financed proceeds at shipment (day 60)	Credit-financed proceeds at shipment (day 60)
GROSS AMOUNT RECEIVED	\$570,000	\$600,000	\$600,000	\$600,000	\$600,000
Bank charge: LC advising	\$0	\$150	\$150	\$0	\$0
Bank charge: LC confirmation (0.15%/month × months × LC value)	\$0	\$1,800	\$0	\$0	\$0
Bank charge: negotiation / handling (0.20% × bill)	\$0	\$1,200	\$1,200	\$0	\$0
Bank charge: D/A collection handling (0.15% × bill)	\$0	\$0	\$0	\$900	\$0
Bank charge: open account remittance handling	\$0	\$0	\$0	\$0	\$150
Bank charge: inward remittance (advance)	\$100	\$0	\$0	\$0	\$0
Bank charge: SWIFT / courier	\$0	\$120	\$120	\$120	\$120
TOTAL BANK CHARGES	\$100	\$3,270	\$1,470	\$1,020	\$270
Financing / discounting cost (usance LC: bill × 6.5% × 90/360; D/A & OA: bill ×	\$0	\$0	\$9,750	\$9,750	\$9,750
Packing credit cost (Task A)	\$0	\$4,200	\$4,200	\$4,200	\$4,200
Default probability	0.0%	0.2%	1.0%	4.0%	8.0%
Expected loss = probability × order value	\$0	\$1,200	\$6,000	\$24,000	\$48,000
NET REALISATION (nominal) = gross – charges – financing – packing cred	\$569,900	\$591,330	\$578,580	\$561,030	\$537,780
Net realisation, INR (× 83.50)	₹47,586,650	₹49,376,055	₹48,311,430	₹46,846,005	₹44,904,630
Days from day 0 to cash	0	60	60	60	60
PV factor (simple) = 1 / (1 + 12% × days/360)	1.000000	0.980392	0.980392	0.980392	0.980392
PRESENT VALUE OF NET REALISATION, USD	\$569,900	\$579,735	\$567,235	\$550,029	\$527,235
Present value, INR	₹47,586,650	₹48,407,897	₹47,364,147	₹45,927,456	₹44,024,147
ALL-IN COST % = (order value – PV) / order value	5.02%	3.38%	5.46%	8.33%	12.13%
All-in cost, USD (order value – PV)	\$30,100	\$20,265	\$32,765	\$49,971	\$72,765
Rank by PV of net realisation (1 = best)	2	1	3	4	5
Rank by default risk (1 = safest)	1	2	3	4	5
Cost of risk share: expected loss as % of all-in cost	0.0%	5.9%	18.3%	48.0%	66.0%

Checks

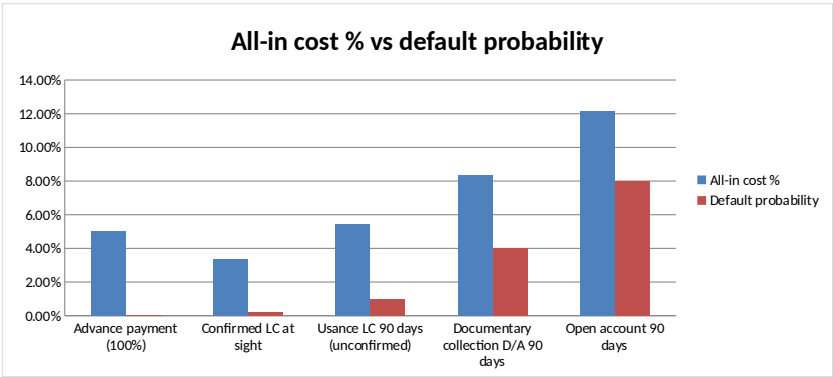
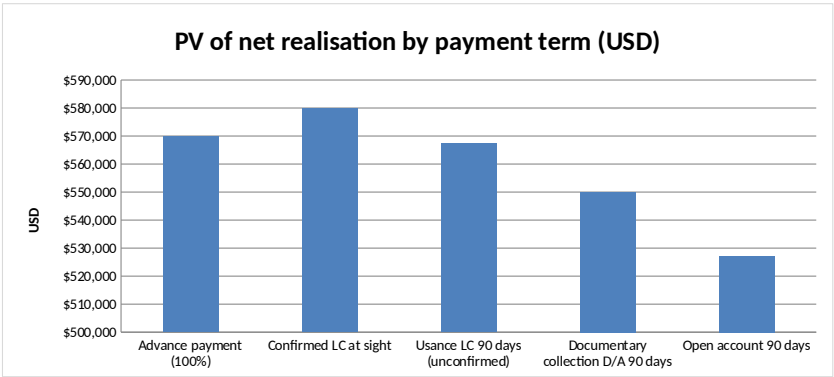
Advance premium check: does the 5% concession (USD) exceed total sight-LC	\$30,000	\$8,670	YES — concession costs more than sight LC
Best term by PV (formula)	Confirmed LC at sight		
Safest term by default risk (formula)	Advance payment (100%)		
PV gap: best term minus second-best, USD	\$9,835		

TASK D — Ranking, comparison table and chart

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Payment term	Default probability	Expected loss (USD)	Net realisation nominal (USD)	PV of net realisation (USD)	PV (INR)	All-in cost %	Rank by PV	Rank by risk	Verdict
Advance payment (100%)	0.0%	\$0	\$569,900	\$569,900	₹47,586,650	5.02%	2	1	Runner-up
Confirmed LC at sight	0.2%	\$1,200	\$591,330	\$579,735	₹48,407,897	3.38%	1	2	RECOMMENDED
Usance LC 90 days (unconfirmed)	1.0%	\$6,000	\$578,580	\$567,235	₹47,364,147	5.46%	3	3	Acceptable fallback
Documentary collection D/A 90 days	4.0%	\$24,000	\$561,030	\$550,029	₹45,927,456	8.33%	4	4	Reject — risk too high
Open account 90 days	8.0%	\$48,000	\$537,780	\$527,235	₹44,024,147	12.13%	5	5	Reject — risk too high

Recommended structure: Confirmed LC at sight
Advance PV shortfall vs recommended, USD: \$9,835
Open account PV shortfall vs recommended, U \$52,500
Is advance payment really best once the conceNo — the 5% concession (USD 30,000) costs more than every friction on the sight LC combined



TASK E — UCP 600 documentary check on the confirmed sight LC presentation and its financial impact

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#	Document / item presented	LC requirement	Presented as	Discrepancy?	UCP 600 basis	Why it fails
1	Commercial invoice	Signed invoice for the LC amount (USD 600,000)	Shows USD 605,000	YES	Art. 18(b) — a nominated/confirming bank may refuse an invoice issued for an amount in excess of the credit; Art. 14(d) data must not conflict	Invoice exceeds the LC amount by USD 5,000; no ±tolerance wording ("about"/"approximately") in the LC (Art. 30).
2	Ocean bill of lading	Full set of CLEAN SHIPPED ON BOARD B/L	Marked "received for shipment"	YES	Art. 20(a)(ii) — B/L must indicate goods shipped on board a named vessel by pre-printed wording or an on-board notation with date	"Received for shipment" only proves receipt by the carrier, not loading; no on-board date, so shipment cannot be evidenced.
3	Marine insurance policy	Cover for 110% of CIF value	Covers 100% of value	YES	Art. 28(f)(ii) — minimum cover is 110% of CIF/CIP value when the credit so requires (and even by default)	Under-insured by 10 percentage points; the bank/buyer are not protected for the full landed value plus expected profit.
4	Certificate of origin	Consistent with other documents	Consistent	NO	Art. 14(d)(e) — data need not be identical but must not conflict	Complies — no action.
5	Shipment date	On or before the LC's latest shipment date	Later than the latest shipment date	YES	Art. 14(c) and the LC's latest-shipment field (MT700 44C); late shipment is a classic discrepancy	Documents dated after the latest shipment date cannot be honoured; also risks the Art. 14(c) 21-day presentation window and LC expiry.
6	Packing list & sight bill of exchange	Required	Presented (no issue stated)	NO	Art. 14	Complies — no action.

Number of discrepancies found 4
Discrepancy fee, USD (USD 90 × se \$90

FINANCIAL IMPACT ON THE SIGHT LC OPTION

Line	Compliant presentation	Discrepant presentation	Change	Explanation
Default probability	0.2%	8.0%	7.8%	Bank undertaking is lost (Art. 16 refusal) — payment now depends on the buyer's waiver, i.e. the buyer's own credit risk (open-account level). Probability × exposure.
Expected loss, USD	\$1,200	\$48,000	\$46,800	
Discrepancy fee, USD	\$0	\$90	\$90	USD 90 per set presented.
Days to cash	60	70	10	Waiver / re-presentation delay (scenario input).
Net realisation (nominal), USD	\$591,330	\$544,440	-\$46,890	Extra expected loss and fee reduce the nominal figure.
PV factor	0.980392	0.977199	-0.003193	Longer wait lowers the PV factor.
PV of net realisation, USD	\$579,735	\$532,026	-\$47,709	Combined effect.
All-in cost %	3.38%	11.33%	7.95%	Sight LC drifts from best option to open-account-like economics.
Rank among the five terms (PV)	1	4	3	Recomputed rank of the discrepant LC against the other four terms.

Cure actions: (1) re-issue invoice at USD 600,000; (2) obtain on-board notation with date on the B/L; (3) endorse insurance to 110% CIF; (5) request LC amendment extending latest shipment date and expiry BEFORE presenting — then the presentation is compliant and the confirming bank's undertaking is restored.

SENSITIVITY — how the ranking moves with cost of capital and open-account default risk (all live formulas)

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PV of net realisation (USD) at different exporter costs of capital						
Payment term	8%	10%	12%	14%	16%	
Advance payment (100%)	\$569,900	\$569,900	\$569,900	\$569,900	\$569,900	\$569,900
Confirmed LC at sight	\$583,549	\$581,636	\$579,735	\$577,847		\$575,971
Usance LC 90 days (unconfirmed)	\$570,967	\$569,095	\$567,235	\$565,388		\$563,552
Documentary collection D/A 90 days	\$553,648	\$551,833	\$550,029	\$548,238		\$546,458
Open account 90 days	\$530,704	\$528,964	\$527,235	\$525,518		\$523,812

Best term at each rateConfirmed LC at sightConfirmed LC at sightConfirmed LC at sightConfirmed LC at sightConfirmed LC at sight

Break-even default probability at which open account would equal the sight LC on PV

OA net realisation before expected loss, USD	\$585,780
Required OA net realisation to match sight-LC PV, USD	\$591,330
Break-even default probability	-0.93%
Interpretation	Open account can never beat the sight LC even at zero default risk
Advance concession break-even: concession % at which advance payment equals sight LC	3.36%
Interpretation	Advance beats the sight LC only if the buyer's concession is below 3.36% (buyer demands 5.00%)